

Precision-Retention Frontiers for Low-Precision Analog Vision via Ensemble Hardware-Aware Training

Songqiao Li

Abstract

Analog compute-in-memory (CIM) can reduce movement-dominated inference cost, but deployment depends on two separable questions: whether a trained network transfers across hardware instances and whether the programmed memory state remains accurate over time. We present a behavioral simulation framework that decouples these questions for low-precision vision transformers. In an idealized AIHWKit IdealDevice ablation, standard fixed-instance hardware-aware training reaches 87.28% fresh-instance accuracy at 8-bit precision but falls to 14.64% at 4-bit; Ensemble Hardware-Aware Training (HAT), which resamples device-to-device (D2D) mismatch masks during training, restores the 4-bit pure-quantization ablation to 86.16%. Reintroducing realistic phase-change-memory (PCM) UnitCell physics produces a different precision-retention frontier: 8-bit PCM gives the highest fresh accuracy (77.60%) with negligible drift, 4-bit PCM remains competitive immediately after programming (76.68%) but violates a 1 pp deployment-drift SLA after a 4.01 pp one-day loss, and 6-bit PCM enters a D2D-sensitive transition zone with lower mean accuracy (68.44%, seed-level std 6.03%) despite drift stability (<0.1 pp). Thus the algorithmic result is not itself a PCM deployment claim. Ensemble HAT addresses cross-instance robustness, whereas the deployable PCM operating point is governed by retention and quantization–D2D interactions.

Keywords: analog compute-in-memory; hardware-aware training; phase-change memory; low-precision inference; precision-retention tradeoff; vision transformers; mixed-signal simulation.

1 Introduction

Compute-in-memory (CIM) architectures execute dense matrix-vector multiplications inside memory arrays, reducing the data movement that dominates conventional digital inference^{1–3}. The same physical proximity that makes CIM attractive also exposes neural networks to device-level constraints: finite conductance precision, device-to-device variability, cycle-to-cycle read noise, and retention drift^{4–8}. These effects are often measured or modeled at the device level, but their system-level consequences for modern vision backbones remain difficult to rank before fabrication.

The central difficulty is diagnostic. Different failure modes can produce the same accuracy loss: a low-precision model may overfit one sampled mismatch realization during hardware-aware training, lose information to a coarse conductance grid, or drift after programming. Mature CIM simulators expose increasingly detailed non-idealities^{2;4;5;9}, yet device-specific studies, including organic and optoelectronic demonstrations, often treat variability, ADC resolution, retention-induced drift, and transfer across fabricated instances qualitatively or in isolation^{10–12}. A deployment workflow therefore needs to distinguish an algorithmic robustness problem from a physical precision-retention limit.

Here we present a behavioral simulation framework that evaluates analog neural network deployment under both idealized pure-quantization and realistic device-physics regimes. We apply a mixed-signal deployment model to the Tiny-ViT-5M backbone and use the contrast between these regimes to separate cross-instance generalization from physical retention limits. The paper makes three contributions. First, it maps a 4/6/8-bit precision-retention frontier for PCM and finds a non-monotonic precision–accuracy curve. In this tested sweep, 8-bit PCM is drift-flat with the highest fresh accuracy (77.60%), 4-bit PCM is trainable but drift-limited, and 6-bit PCM enters a D2D-sensitive transition zone with lower mean accuracy (68.44%, seed-level std 6.03%) despite drift stability. This behavior is consistent with a quantization–D2D

interaction in which mismatch perturbations become comparable to the 1/64 conductance step. Second, the study explains why this frontier cannot be predicted from algorithm-only tests: in an idealized pure-quantization regime, standard fixed-instance hardware-aware training at 4-bit collapses to near-chance, while Ensemble Hardware-Aware Training (HAT) restores fresh-instance accuracy to $86.16 \pm 0.19\%$ by resampling D2D mismatch masks each epoch. Third, the comparison yields a deployment rule: solve cross-instance robustness first, then choose the memory precision and readout regime that satisfy retention and variability constraints.

The framework is intentionally behavioral rather than circuit-accurate. It focuses on macroscopic precision, mismatch, readout, and drift phenomena, while leaving array-level parasitics such as IR drop and sneak paths to future circuit-calibrated studies. Additional supporting analyses, including front-end compensation and zero-shot transfer to alternative organic optoelectronic profiles, are provided in the Supplementary Information. The goal is not to replace detailed circuit simulators, but to provide a reproducible layer for comparing mitigation strategies and identifying deployment frontiers before hardware-specific calibration is available.

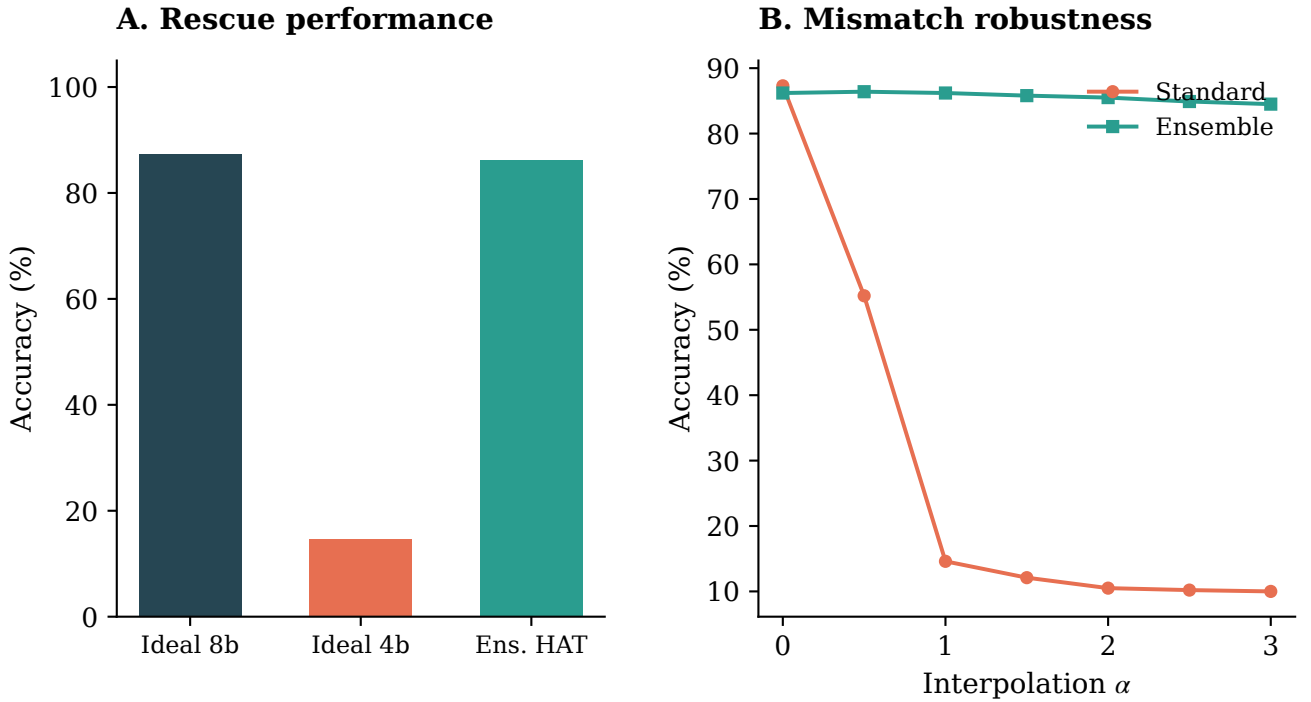


Figure 1: Algorithmic generalization failure in the ideal-device ablation. (A) Fresh-instance accuracy for the AIHWKit IdealDevice baseline. 4-bit pure quantization falls to near chance, while Ensemble HAT restores performance near the 8-bit baseline. This is an algorithmic ablation, not the PCM deployment headline. (B) Loss-landscape interpolation from the source training mask ($\alpha = 0$) to fresh hardware instances ($\alpha = 1$). Ensemble HAT discovers a broad plateau robust to D2D mismatch, while Standard HAT exhibits extreme sensitivity.

2 Related Work

Hardware-aware training (HAT) mitigates analog non-idealities by injecting hardware perturbations into the forward path during optimization¹³. It is related to quantization-aware training¹⁴ and to domain randomization in sim-to-real transfer¹⁵, but CIM introduces a distinct transfer problem: device-to-device (D2D) mismatch is spatially structured and fixed for a fabricated instance, whereas cycle-to-cycle (C2C) noise is stochastic per read. Training against one fixed mismatch field can therefore produce an instance-specific solution that does not transfer. Ensemble HAT applies the domain-randomization idea at the hardware-instance level by resampling the static D2D mask across epochs.

The need for this distinction is especially visible in organic neuromorphic and optoelectronic devices.

These devices offer optical sensitivity, multilevel conductance tuning, and flexible-substrate compatibility for neuromorphic hardware^{6;8;16;17}. Recent demonstrations report multilevel memory^{7;11}, long retention tails¹⁰, and integrated optoelectronic arrays^{18–21}. The system question is now becoming more explicit: active-matrix addressing, spatial optical response, crosstalk, and temperature stability are increasingly treated as deployment constraints rather than only as materials observations^{22–26}. This motivates network-level tools that can substitute device profiles and rank failure modes before a full array is available.

Existing CIM simulators provide the foundation for such studies. DNN+NeuroSim established an effective template for connecting device assumptions, peripheral-circuit costs, and neural-network accuracy². MemTorch embeds memristive non-idealities into PyTorch workflows for end-to-end evaluation⁹, AIHWKit provides analog HAT and inference simulation in a practical mixed-signal software stack⁴, and CrossSim supplies a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models⁵. These mature simulators remain the correct tools for circuit-aware studies, but they do not expose every profile-level effect of interest, such as inverse-gamma optoelectronic photoresponse or organic-specific double-exponential retention, as first-class device primitives. Our behavioral profile-driven layer is therefore positioned as a deployment-risk ranking extension, not as a replacement for established CIM toolkits.

3 Methodology

The methodology is designed to make attribution possible. Across the main experiments, the network, analog/digital mapping, and optimization recipe are held fixed while the perturbation model is changed. A performance change can therefore be interpreted as evidence for either an algorithmic transfer limit across hardware instances or a physical deployment constraint such as precision, drift, or readout resolution.

The first component is a hybrid analog/digital inference stack in which dense linear operators execute on differential-pair crossbars and control-heavy operators remain digital. The analog partition contains patch embeddings, query/key/value (Q/K/V) and output projections, and feed-forward layers. The digital partition retains softmax, layer normalization, GELU activation, positional encoding, residual additions, and the class token, following established practice in heterogeneous CIM accelerators^{27–31}.

Under this mapping, approximately 88% of parameters execute on analog arrays while roughly 60% of estimated energy remains digital, largely because attention operations stay in the digital domain. The split is therefore not a claim that all transformer computation becomes analog; it is a controlled mixed-signal deployment point consistent with recent heterogeneous CIM accelerators for Vision Transformers^{28–31}.

Table 1: Canonical Simulation Parameters and Hardware Constraints. These values define the default non-ideality regime for the Tiny-ViT-5M experiments unless otherwise specified.

Parameter	Value	Physical Meaning
n_{states}	16	Conductance resolution (4-bit equivalent)
σ_{D2D}	10%	Device-to-device mismatch (relative std. dev.)
σ_{C2C}	5%	Cycle-to-cycle read noise
$G_{\text{max}}/G_{\text{min}}$	10	Dynamic range of analog weights
ADC bits	8-bit	Default readout quantization (before sweep)

The physical model then injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and nonlinear-write surrogates directly into the forward path. Each analog-mapped weight is decomposed into positive and negative branches, mapped to the conductance window $[G_{\text{min}}, G_{\text{max}}]$, and quantized to n_{states} levels via a straight-through estimator (STE) quantizer $Q_n(\cdot)$. Differential readout recovers $G_{\text{eff}} = \tilde{G}^+ - \tilde{G}^-$, giving the downstream digital weight

$$\hat{W}_{ij} = s_{\ell} G_{\text{eff},ij}, \quad s_{\ell} = \begin{cases} \frac{w_{\text{max}}}{G_{\text{max}} - G_{\text{min}}}, & \text{standard calibration} \\ \frac{w_{\text{max}}}{\max_{i,j} |G_{\text{eff},ij}| + \epsilon}, & \text{retention-time recalibration} \end{cases} \quad (1)$$

with layer index ℓ omitted. Unless otherwise stated, we use the standard-calibration branch; the retention-recalibration branch is used only for specific retention-drift diagnostics. The recovered weight step is $\Delta_{\hat{w}} = s_\ell(G_{\max} - G_{\min})/(n_{\text{states}} - 1)$, so perturbations smaller than half a step are masked. The differential-pair scheme suppresses common-mode noise^{32;33}.

Hardware-aware training (HAT) injects forward perturbations while back-propagating through a straight-through estimator. Standard HAT fixes one D2D mismatch field M_0 during training. That objective is appropriate when deployment reuses the same calibrated instance, but it is a weak proxy for zero-shot transfer to a fresh hardware draw. Ensemble HAT instead treats the mismatch map as a random variable resampled each epoch:

$$\theta_{\text{ens}}^* = \arg \min_{\theta} \mathbb{E}_{M \sim \mathcal{D}_{\text{D2D}}} \left[\frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M), y_n) \right], \quad (2)$$

where the expectation is Monte-Carlo estimated with one fresh draw $M^{(e)} \sim \mathcal{D}_{\text{D2D}}$ held constant across all mini-batches in epoch e . Expanding Eq. 2 (Supplementary Note S-Theory) reveals an implicit Fisher-weighted gradient-squared regularization effect that pushes the optimizer toward flat regions in the M -direction. Epoch-level, rather than per-batch, resampling is important: it gives the optimizer a consistent instance long enough to learn from it before the next mismatch field is introduced¹⁵. Fresh-instance transfer is evaluated on unseen i.i.d. D2D draws held constant for each evaluation pass. For a single checkpoint, the diagnostic spread is the standard deviation across ten fresh-instance means (each averaged over five forward passes); for the primary results, the uncertainty is the sample standard deviation across three seed-level fresh-instance means¹.

Algorithm 1 Ensemble Hardware-Aware Training (HAT)

```

1: procedure TRAIN( $\mathcal{D}_{\text{train}}, \eta, \mathcal{D}_{\text{D2D}}$ )
2:   Initialize weights  $\theta$ 
3:   for epoch  $e = 1 \dots E$  do
4:     Sample fresh mismatch mask  $M^{(e)} \sim \mathcal{D}_{\text{D2D}}$ 
5:     for batch  $(x, y) \in \mathcal{D}_{\text{train}}$  do
6:       Compute output with current mask:  $\hat{y} = f(x; \theta, M^{(e)})$ 
7:       Update  $\theta \leftarrow \theta - \eta \nabla_{\theta} \ell(\hat{y}, y)$ 
8:     end for
9:   end for
10:  return  $\theta$ 
11: end procedure
12: procedure EVALUATE( $x, \theta, \mathcal{D}_{\text{D2D}}$ )
13:  Sample unseen masks  $\{M'_k\}_{k=1}^K \sim \mathcal{D}_{\text{D2D}}$ 
14:  return  $\frac{1}{K} \sum_{k=1}^K \text{Acc}(f(x; \theta, M'_k))$ 
15: end procedure

```

All Tiny-ViT experiments then use a fixed training recipe so that changes in accuracy can be attributed to the hardware model rather than to optimizer retuning. We use AdamW ($lr = 5 \times 10^{-4}$, weight decay 0.05), 100 epochs, batch size 64, and cosine annealing ($T_{\max} = 100$). Images are resized to 224×224 with standard augmentation. The default analog resolution is $n_{\text{states}} = 16$ levels. The core comparison uses four configurations: a digital reference, a hybrid baseline without analog noise, a fixed-mismatch noisy baseline, and Ensemble HAT. Additional variants introduce front-end models and retention-drift corrections only for diagnostic checks (see Supplementary Note S-Notation).

Nonlinear-write and retention surrogates are anchored to measured DNTT transients³⁴. Nonlinear write modifies the STE backward pass by scaling the straight-through gradient with a state-dependent

¹Single-checkpoint diagnostics are reported only as provenance in the Supplementary Information. The main claim uses the 3-seed mean $86.16 \pm 0.19\%$ to emphasize reproducibility robustness.

factor proportional to the present conductance magnitude raised to $NL - 1$ ²; ideal write is recovered when $NL_{LTP} = 1$ (long-term potentiation) and $|NL_{LTD}| = 1$ (long-term depression). The optional optoelectronic front-end model applies inverse-gamma preprocessing, photocurrent shot-noise injection, and per-sample renormalization; the explicit response equations are kept in the Supplementary Information.

To make later device substitution possible, the profile interface packages technology-specific assumptions into a replaceable JSON parameter bundle. Deployment holds the digital backbone and training recipe fixed while substituting the profile, allowing literature priors to support design-space ranking today and measured-device calibration to replace them later without rewriting the training stack.

For the joint D2D–ADC sweep, first-order factor importance is quantified with Sobol indices estimated from the 7×9 grid of Monte Carlo means. Energy is estimated via a first-order analytical model (Supplementary Methods) from operator counts and literature-proxy constants ($E_{\text{cell}} = 100$ fJ, $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ). These values support relative comparisons only; they are not measured silicon energy numbers.

4 Experimental Setup

We evaluate ResNet-18, ConvNeXt-Tiny, and Tiny-ViT-5M backbones across CIFAR-10, CIFAR-100, and Flowers-102. This set provides a progression from entry-level convolutional validation to a compact transformer deployment target. The digital FP32 baselines are summarized in Table 2. Models for CIFAR-10/100 were trained from scratch, while Flowers-102 used ImageNet-pretrained weights.

Table 2: Evaluated backbones and digital (FP32) accuracy baselines. These baselines define the ideal software performance before analog non-ideality injection.

Dataset	ResNet-18	ConvNeXt-Tiny	Tiny-ViT-5M
CIFAR-10	95.46%	90.74%	98.06%
CIFAR-100	78.64%	64.12%	86.94%
Flowers-102	n.a.	33.22%	97.97%

The main text then focuses on Tiny-ViT-5M under two regimes: an extreme low-precision 4-bit ablation used to diagnose cross-instance transfer, and a realistic phase-change memory (PCM) precision ladder spanning 4/6/8-bit operation. Detailed optimization settings, Monte Carlo evaluation protocols, and auxiliary architectural experiments, including ConvNeXt retention and proportional-noise studies, are summarized in the Supplementary Information.

Parameter provenance is summarized in the Supplementary Information together with the optimization settings and evaluation conditions used for the reported runs. The study targets reproducible reruns through archived configurations, logs, and model lineage. Because the framework operates at the simulation level, device parameters are literature-derived or proxy-calibrated rather than extracted from fabricated devices under direct measurement. The quantitative results should therefore be interpreted as comparative risk rankings and deployment frontiers for the tested profiles, not as absolute hardware predictions.

5 Results

The results are organized to avoid conflating two different questions. We first ask whether a low-precision model trained with analog perturbations transfers to a fresh hardware instance. We then reintroduce PCM device physics to determine which precision regime remains deployable after that transfer problem is separated from retention and readout limits.

²The absence of an explicit NL prefactor is intentional: this is a behavioral proxy for position-dependent update difficulty, not the strict derivative of a power-law $f(G) = G^{NL}$.

5.1 IdealDevice Ablation: Fixed-Mask Failure and Algorithmic Rescue

We first use the tested AIHWKit IdealDevice quantization/noise baseline as an algorithmic ablation, not as the physical PCM deployment result. The fixed-mask HAT protocol mirrors the common practice of optimizing against one sampled hardware instance or analog tile configuration^{4,13}. This setting is useful because it exposes the boundary of instance-specific training. As summarized in Table 3, standard per-batch hardware-aware training reaches 87.28% fresh-instance accuracy at 8-bit precision, but falls to 14.64% at 4-bit (Figure 1A). The failure is therefore not ordinary optimization failure; it is a transfer failure caused by fresh hardware instance shifts at extreme low precision.

Table 3: IdealDevice pure-quantization ablation. Data show fresh-instance accuracy across three seeds for the AIHWKit IdealDevice model. These results establish algorithmic cross-instance robustness, not PCM physical deployment performance.

Model Configuration	Precision	Fresh Accuracy	Role
IdealDevice Baseline	8-bit	$87.28 \pm 0.13\%$	Stable upper bound
IdealDevice Baseline	4-bit	$14.64 \pm 0.11\%$	Generalization collapse
Ensemble HAT	4-bit	$86.16 \pm 0.19\%$	Algorithmic rescue

To localize the error source, we remove constraints sequentially (Table 4). Quantization and hybrid mapping alone introduce only a small penalty ($98.06\% \rightarrow 97.39\%$). The collapse appears when a fixed mismatch mask is introduced, indicating that the model has learned a solution tied to one hardware draw rather than a solution that transfers across the mismatch distribution. Ensemble HAT mitigates this failure by forcing the optimizer to see a sequence of mismatch fields during training.

Table 4: Ablation of mapping and noise constraints. Accuracy reported for the Tiny-ViT-5M backbone at 4-bit precision (where applicable) on CIFAR-10.

Configuration	Precision	Partition	Mismatch Mask	Accuracy
Digital Reference	32-bit	Digital	None	$98.06 \pm 0.14\%$
Hybrid Baseline	4-bit	Analog/Digital	None	$97.39 \pm 0.08\%$
Fixed-mask HAT	4-bit	Analog/Digital	Single (fixed)	$10.00 \pm 0.00\%$
Ensemble HAT	4-bit	Analog/Digital	Resampled	$86.16 \pm 0.19\%$

By resampling the device-to-device (D2D) variation pattern during training (Figure 1A), Ensemble HAT restores this 4-bit pure-quantization ablation to 86.16% fresh-instance accuracy. The interpretation is deliberately narrow: 4-bit operation can be algorithmically viable when device physics are idealized, but this does not imply that 4-bit PCM remains stable after programming. Loss-landscape interpolation along the D2D mismatch direction (Figure 1B) supports this interpretation. Ensemble HAT finds a broad plateau in the mismatch direction, whereas Standard HAT converges to a sharp, instance-specific minimum.

5.2 Realistic PCM Training and the Precision-Retention Frontier

Having isolated the algorithmic transfer problem, we next ask whether a realistic memory model preserves the same low-precision promise. We evaluate the trained deployment with the PCMPresetUnitCell model in AIHWKit, which includes nonlinear update physics, finite dynamic range, and time-dependent conductance drift.

Under the tested PCM UnitCell simulation regime, training remains viable at 4-bit, 6-bit, and 8-bit precision, but viability is not monotonic in bit width. We mark a setting as drift-constrained when its 24-hour accuracy loss exceeds a 1 pp deployment SLA. By this criterion, 8-bit and 6-bit PCM are effectively drift-flat, whereas 4-bit PCM incurs a 4.01 pp retention penalty and is therefore drift-limited (Table 5). The important exception to a simple “more bits is better” reading is the 6-bit row: it is drift-stable, yet it has the lowest mean fresh accuracy. This behavior is consistent with a quantization-D2D interaction

in which mismatch perturbations become comparable to the 1/64 conductance step. At 4-bit, coarser quantization masks part of the D2D perturbation; at 8-bit, the finer grid absorbs it; at 6-bit, the two effects interact unfavorably in this tested regime.

Table 5: PCM Precision Ladder and Stability Frontier. Fresh accuracy reflects corrected new-protocol evaluations (3 seeds for 4/8-bit; 4 seeds for 6-bit). The retention columns are from the drift-evaluation protocol; Δ drift is computed as retention-eval 0 s accuracy minus 24 h accuracy, not as fresh-eval mean minus 24 h accuracy. The 6-bit row uses the completed four-seed drift closure and shows drift stability but low mean fresh accuracy, consistent with a D2D-sensitive transition regime rather than a deployment frontier optimum.

Precision	Fresh Acc.	1 h Acc.	24 h Acc.	$\Delta_{0 \rightarrow 24h}$	Deployment Role
8-bit PCM	77.60%	77.49%	77.57%	0.04 pp	Drift-flat reference
6-bit PCM	68.44%	68.46%	68.36%	0.04 pp	D2D-sensitive transition zone
4-bit PCM	76.68%	74.04%	72.64%	4.01 pp	Drift-limited regime

5.3 Iso-Accuracy Operating Envelope

The same separation appears at the readout level. In the joint ADC-precision and D2D-variability sweep (Figure 2), accuracy is near chance below 5-bit ADC, and moving from 5 to 6 bits produces a consistent ~ 7 pp gain. Once readout precision reaches this threshold, the limiting factor changes: further degradation is governed mainly by D2D variability rather than ADC quantization. The Sobol analysis captures this hierarchy, with $S_{\text{ADC}} = 0.98$ over the full grid and $S_{\text{D2D}} = 0.92$ after restricting to the operational regime. The resulting hardware prescription is sequential rather than simultaneous: secure 6-bit readout first, then spend device-engineering effort on mismatch reduction.

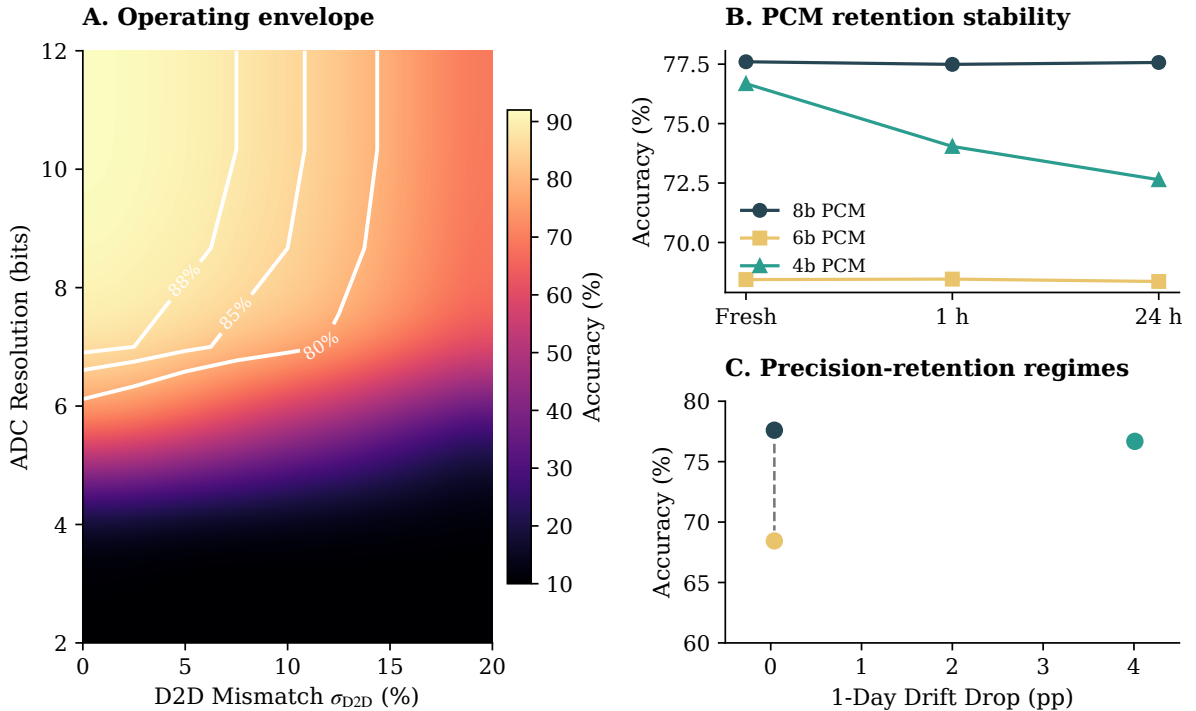


Figure 2: Physical constraints and the iso-accuracy operating envelope. Contour map under joint ADC precision and D2D mismatch sweep. Readout precision forms a hard cliff below 6 bits, above which D2D variability dictates the operational boundary. Bold contour lines indicate the 80%, 85%, and 88% accuracy levels.

The operating envelope turns the ablations into a deployment sequence rather than a single best training recipe. First, cross-instance overfitting must be removed by training against a distribution of mismatch fields. Second, the readout interface must clear the 6-bit ADC cliff. Third, the memory

precision must be selected under the target retention constraint. Under the tested PCM model, 8-bit PCM is the strongest operating point because it combines 77.60% fresh accuracy with negligible drift. 4-bit PCM remains competitive immediately after programming (76.68%) but loses ~ 4.01 pp over one day and violates the 1 pp drift SLA. 6-bit PCM is not drift-limited, but its 68.44% fresh accuracy and high seed-to-seed variance (std 6.03%) mark it as a D2D-sensitive transition zone rather than a deployment optimum.

6 Discussion

The main implication is not that one analog precision is universally optimal. It is that analog deployment has to be staged: first prevent the network from overfitting a single hardware instance, then choose readout and memory precision under the target device physics.

The first deployment frontier is algorithmic. Standard fixed-mask hardware-aware training can be effective when precision is sufficient or when the trained hardware instance is reused, but it becomes fragile at extreme low precision because the optimizer can exploit one sampled mismatch field. The model then performs well for the training instance and poorly for fresh hardware draws. Ensemble HAT changes the objective: by exposing the model to a distribution of mismatch fields during training, it discourages instance-specific solutions and makes fresh-instance transfer part of the optimization target.

That algorithmic rescue does not remove the device frontier; it makes the frontier measurable. Once cross-instance overfitting is controlled, the limiting question becomes whether the memory substrate maintains a usable state after programming. In the tested PCM UnitCell models, 8-bit precision is the strongest practical operating point because it combines high fresh accuracy (77.60%) with drift stability. The 6-bit result is more informative than a monotonic scaling law would be: it is drift-stable, but D2D perturbations reduce mean accuracy (68.44%) and increase seed-to-seed variance (std 6.03%). It therefore marks a transition zone between the quantization-dominated 4-bit regime and the noise-absorbed 8-bit regime.

The joint ADC–D2D sweep gives a similarly ordered hardware roadmap. ADC resolution is a hard gate: below 6 bits, classification accuracy collapses regardless of device quality. Above that threshold, device-to-device variability becomes the dominant accuracy limiter. This ordering matters for hardware co-design. Improving device uniformity before the readout interface clears the ADC cliff will not recover the model; after the cliff is cleared, mismatch reduction becomes the correct target.

The conclusions are bounded by the present behavioral simulation stack. Device profiles are calibrated from literature values or a single fabrication batch, so the numbers should not be read as cross-batch, temperature, aging, or circuit-layout coverage. We model quantization, D2D/C2C variability, ADC precision, write nonlinearity, and retention, but defer parasitic IR drop, sneak paths, and pulse-faithful programming to follow-on hardware-calibrated studies.

The energy estimates are likewise first-order analytical projections from literature-proxy constants ($E_{\text{cell}} = 100$ fJ, $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ; Methods). They rank deployment risks rather than predict silicon performance. The current ablation matrix also prioritizes non-ideality coverage over dataset breadth; cross-architecture TinyImageNet sweeps (see Supplementary Information) are treated as validation rather than as main-text claims.

Table 6: Deployment Risk Ranking and Mitigation Strategies. Factors are ranked by their impact on categorical accuracy as identified by Sobol sensitivity analysis and precision-ladder sweeps.

Factor	Impact Rank	Mitigation Strategy
ADC Resolution	High (Hard Gate)	Secure ≥ 6 -bit readout precision
Stochastic D2D	High (Generalization)	Ensemble HAT (distribution matching)
PCM Drift	Moderate (Time-bound)	8-bit for stability; refresh 4-bit periodically
Write Nonlinearity	Moderate (Optimization)	STE gradient scaling
Shot Noise	Low (Transduction)	Front-end inverse-gamma compensation

The next tests are straightforward: replace UnitCell presets with measured-array statistics, verify whether the non-monotonic precision-accuracy curve and 6-bit transition-zone behavior persist under measured devices, and evaluate memory-bound inference components such as analog KV-cache with explicit train/eval noise separation. These extensions do not alter the present conclusion: algorithmic cross-instance resampling and physical precision-retention selection are separate deployment decisions.

7 Conclusion

We presented a behavioral simulation framework for analog compute-in-memory inference that separates algorithmic cross-instance transfer from physical precision-retention constraints. By comparing ideal pure-quantization baselines with realistic phase-change memory (PCM) physics, the framework identifies deployment frontiers before committing to a specific device implementation.

Two conclusions are central. First, the tested PCM UnitCell regime exposes a non-monotonic precision-accuracy curve: 8-bit PCM achieves the highest fresh accuracy (77.60%) with negligible drift, 4-bit PCM achieves comparable fresh accuracy (76.68%) but pays a 4.01 pp one-day drift penalty, and 6-bit PCM enters a D2D-sensitive transition zone with lower mean accuracy (68.44%, seed-level std 6.03%) despite drift stability. Second, the ideal-device ablation shows why this physical frontier should not be confused with algorithmic robustness. Fixed-instance training at 4-bit collapses to 14.64%, whereas Ensemble Hardware-Aware Training (HAT) resamples mismatch masks and recovers fresh-instance accuracy to $86.16 \pm 0.19\%$.

The resulting design rule is sequential. A system designer should first avoid fixed-instance overfitting, then clear the readout-precision cliff, and only then select the memory precision that satisfies retention constraints. Ensemble HAT addresses the first step; the ADC/D2D envelope and PCM precision ladder define the hardware-side gates. The same separation should guide future studies of memory-bound inference components, including analog KV-cache architectures, where train-time robustness and deployment-time device physics must again be evaluated independently.

References

- [1] Mark Horowitz. Computing’s energy problem (and what we can do about it). In *IEEE International Solid-State Circuits Conference (ISSCC)*, pages 10–14, 2014. doi: 10.1109/ISSCC.2014.6757323. URL <https://doi.org/10.1109/ISSCC.2014.6757323>.
- [2] Xiaochen Peng, Shanshi Huang, Hongwu Jiang, Anni Lu, and Shimeng Yu. DNN+NeuroSim v2.0: An end-to-end benchmarking framework for compute-in-memory accelerators for on-chip training. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 40(11): 2306–2319, 2021. doi: 10.1109/TCAD.2020.3043731. URL <https://doi.org/10.1109/TCAD.2020.3043731>.
- [3] Mario Lanza et al. The growing memristor industry. *Nature*, 640:613–622, 2025. doi: 10.1038/s41586-025-08733-5.
- [4] Malte J. Rasch, Diego Moreda, Tayfun Gokmen, Manuel Le Gallo, Fabio Carta, Cindy Goldberg, Kaoutar El Maghraoui, Abu Sebastian, and Vijay Narayanan. A flexible and fast PyTorch toolkit for simulating training and inference on analog crossbar arrays, 2021. URL <https://arxiv.org/abs/2104.02184>.
- [5] Sandia National Laboratories. Crosssim: Sandia’s simulator for analog ai accelerators. Technical report, Sandia National Laboratories, 2024. URL <https://doi.org/10.2172/2585829>.
- [6] Yunchao Xu, Yuan He, Dongyong Shan, Biao Zeng, and Qian-Xi Ni. Emerging artificial synaptic devices based on organic semiconductors: Molecular design, structure and applications. *ACS*

- Applied Materials & Interfaces*, 2025. doi: 10.1021/acsami.4c17455. URL <https://doi.org/10.1021/acsami.4c17455>.
- [7] Haotian Guo, Jing Guo, Yujing Wang, Hezhen Wang, Simin Cheng, Zehao Wang, Qian Miao, and Xiaomin Xu. An organic optoelectronic synapse with multilevel memory enabled by gate modulation. *ACS Applied Materials & Interfaces*, 2024. doi: 10.1021/acsami.3c19624. URL <https://doi.org/10.1021/acsami.3c19624>.
 - [8] Li Zhang, Wei Chen, and Jing Xu. Recent progress in organic optoelectronic synaptic devices. *Photonics*, 12(5):435, 2025. doi: 10.3390/photonics12050435. URL <https://doi.org/10.3390/photonics12050435>.
 - [9] Corey Lammie, Wei Xiang, Bernabé Linares-Barranco, and Mostafa Rahimi Azghadi. Memtorch: An open-source simulation framework for memristive deep learning systems. *Neurocomputing*, 485:124–133, 2022. doi: 10.1016/j.neucom.2022.02.043. URL <https://doi.org/10.1016/j.neucom.2022.02.043>.
 - [10] Jianmin Zeng, Xinhui Chen, Shuzhi Liu, Qilai Chen, and Gang Liu. Organic memristor with synaptic plasticity for neuromorphic computing applications. *Nanomaterials*, 13(5):803, 2023. doi: 10.3390/nano13050803. URL <https://doi.org/10.3390/nano13050803>.
 - [11] Uihoon Jung, Miseong Kim, Jaewon Jang, Jin-Hyuk Bae, In Man Kang, and Sin-Hyung Lee. Formation of cluster-structured metallic filaments in organic memristors for wearable neuromorphic systems with bio-mimetic synaptic weight distributions. *Advanced Science*, 11(9):2307494, 2024. doi: 10.1002/advs.202307494. URL <https://doi.org/10.1002/advs.202307494>.
 - [12] Yu-Pu Lin, Christopher H. Bennett, Théo Cabaret, Damir Vodenicarevic, Djaafar Chabi, Damien Querlioz, Bruno Jousset, Vincent Derycke, and Jacques-Olivier Klein. Physical realization of a supervised learning system built with organic memristive synapses. *Scientific Reports*, 2016. doi: 10.1038/srep31932. URL <https://doi.org/10.1038/srep31932>.
 - [13] Vinay Joshi, Manuel Le Gallo, Simon Haefeli, Irem Boybat, S. R. Nandakumar, Christophe Piveteau, Martino Dazzi, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. Accurate deep neural network inference using computational phase-change memory. *Nature Communications*, 11:2473, 2020. doi: 10.1038/s41467-020-16108-9. URL <https://doi.org/10.1038/s41467-020-16108-9>.
 - [14] Jungwook Choi, Zhuo Wang, Swagath Venkataramani, Pierce I-Jen Chuang, Vijayalakshmi Srinivasan, and Kailash Gopalakrishnan. PACT: Parameterized clipping activation for quantized neural networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1028–1037, 2018. doi: 10.48550/arXiv.1805.06085. URL <http://proceedings.mlr.press/v80/choi18a.html>.
 - [15] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30, 2017. doi: 10.1109/IROS.2017.8202133. URL <https://doi.org/10.1109/IROS.2017.8202133>.
 - [16] Pietro Belleri, Judith Pons i Tarrès, Iain McCulloch, Paul W. M. Blom, Zsolt M. Kovács-Vajna, Paschalis Gkoupidenis, and Fabrizio Torricelli. Unravelling the operation of organic artificial neurons for neuromorphic bioelectronics. *Nature Communications*, 15:5350, 2024. doi: 10.1038/s41467-024-49668-1. URL <https://doi.org/10.1038/s41467-024-49668-1>.
 - [17] Junpeng Ji, Dace Gao, Han-Yan Wu, Miao Xiong, Nevena Stajkovic, Claudia Latte Bovio, Chi-Yuan Yang, Francesca Santoro, Deyu Tu, and Simone Fabiano. Single-transistor organic electrochemical

- neurons. *Nature Communications*, 16:4334, 2025. doi: 10.1038/s41467-025-59587-4. URL <https://doi.org/10.1038/s41467-025-59587-4>.
- [18] Lan Shen Hu, Marco Fattori, Winston Schilp, Roy Verbeek, Setareh Kazemzadeh, Yoeri van de Burgt, Auke Jisk Kronemeijer, Gerwin Gelinck, and Eugenio Cantatore. An energy-efficient solid-state organic device array for neuromorphic computing. *IEEE Transactions on Electron Devices*, 70(12):6520–6525, 2023. doi: 10.1109/TED.2023.3327947. URL <https://doi.org/10.1109/TED.2023.3327947>.
- [19] Xu Liu, Shilei Dai, Yiyang Jin, Junyao Zhang, Ziyi Guo, Tongrui Sun, Li Li, Pu Guo, Huaiyu Gao, Haixia Liang, Shiqi Zhang, Lize Xiong, Yanmin Zhou, and Jia Huang. Near-infrared organic photoelectrochemical synaptic transistors by wafer-scale photolithography for neuromorphic visual system. *Nature Communications*, 17:197, 2026. doi: 10.1038/s41467-025-66891-6. URL <https://doi.org/10.1038/s41467-025-66891-6>. Published online 2025-12-08.
- [20] Fanqing Zhang, Xinyi Shan, Zhenhua Ji, Yan Wang, Jing Zhao, and Zhongrui Wang. Large-scale high uniform optoelectronic synapses array for artificial visual neural network. *Microsystems & Nanoengineering*, 11:42, 2025. doi: 10.1038/s41378-024-00859-2. URL <https://doi.org/10.1038/s41378-024-00859-2>.
- [21] Heyi Huang, Xiangpeng Liang, Yuyan Wang, Jianshi Tang, Yuankun Li, Yiwei Du, Wen Sun, Jianing Zhang, Peng Yao, Xing Mou, Feng Xu, Jinzhi Zhang, Yuyao Lu, Zhengwu Liu, Jianlin Wang, Zhixing Jiang, Ruofei Hu, Ze Wang, Qingtian Zhang, Bin Gao, Xuedong Bai, Lu Fang, Qionghai Dai, Huaxiang Yin, He Qian, and Huaqiang Wu. Fully integrated multi-mode optoelectronic memristor array for diversified in-sensor computing. *Nature Nanotechnology*, 20(1):93–103, 2025. doi: 10.1038/s41565-024-01794-z. URL <https://doi.org/10.1038/s41565-024-01794-z>.
- [22] Ying Liu, Biao Wang, Lei Wu, Lulu Huang, Lu Lin, Li Xiang, Dongqing Liu, Shiguo Zhang, Chenguang Zhu, and Yijie Tao. Artificial visual synaptic architecture with high-linearity light-modulated weight for optoelectronic neuromorphic computing. *ACS Applied Materials & Interfaces*, 2023. doi: 10.1021/acsami.3c11495. URL <https://doi.org/10.1021/acsami.3c11495>.
- [23] Dingwei Li, Yitong Chen, Huihui Ren, Yingjie Tang, Siyu Zhang, Yan Wang, Lixiang Xing, Qi Huang, Lei Meng, and Bowen Zhu. An active-matrix synaptic phototransistor array for in-sensor spectral processing. *Advanced Science*, 11, 2024. doi: 10.1002/advs.202406401. URL <https://doi.org/10.1002/advs.202406401>.
- [24] Tianqing Wan, Bangjie Shao, Sijie Ma, Yue Zhou, Qiao Li, and Yang Chai. In-sensor computing: Materials, devices, and integration technologies. *Advanced Materials*, 35(37):2203830, 2023. doi: 10.1002/adma.202203830. URL <https://doi.org/10.1002/adma.202203830>.
- [25] A. Melianas, T. J. Quill, G. LeCroy, Y. Tuchman, H. v. Loo, S. T. Keene, A. Giovannitti, H. R. Lee, I. P. Maria, I. McCulloch, and A. Salleo. Temperature-resilient solid-state organic artificial synapses for neuromorphic computing. *Science Advances*, 6:eabb2958, 2020. doi: 10.1126/sciadv.abb2958. URL <https://doi.org/10.1126/sciadv.abb2958>.
- [26] Ziyi Guo, Junyao Zhang, Ben Yang, Li Li, Xu Liu, Yutong Xu, Yue Wu, Pu Guo, Tongrui Sun, Shilei Dai, Haixia Liang, Jun Wang, Yidong Zou, Lize Xiong, and Jia Huang. Organic high-temperature synaptic phototransistors for energy-efficient neuromorphic computing. *Advanced Materials*, 36:2310155, 2024. doi: 10.1002/adma.202310155. URL <https://doi.org/10.1002/adma.202310155>.
- [27] Xueying Wu, Edward Hanson, Nansu Wang, Qilin Zheng, Xiaoxuan Yang, Huanrui Yang, Shiyu Li, Feng Cheng, Partha Pratim Pande, and Janardhan Rao Doppa. Block-wise mixed-precision quantization: Enabling high efficiency for practical ReRAM-based DNN accelerators. *IEEE*

- Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2024. doi: 10.1109/TCAD.2024.3409193. URL <https://doi.org/10.1109/TCAD.2024.3409193>.
- [28] Chenyu Wang, Zhen Dong, Daquan Zhou, Zhenhua Zhu, Yu Wang, Jiashi Feng, and Kurt Keutzer. EPIM: Efficient processing-in-memory accelerators based on epitome. In *Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC)*, pages 1–6, 2024. doi: 10.1145/3649329.3657377. URL <https://arxiv.org/abs/2311.07620>.
- [29] Mengke Ge, Junpeng Wang, Binhan Chen, Yingjian Zhong, Haitao Du, Song Chen, and Yi Kang. Allspark: Workload orchestration for visual transformers on processing in-memory systems. *IEEE Transactions on Computers*, 2025. doi: 10.1109/TC.2024.3483633. URL <https://doi.org/10.1109/TC.2024.3483633>.
- [30] Cong Wang, Zexin Fu, Jiayi Huang, and Shanshi Huang. Hemlet: A heterogeneous compute-in-memory chiplet architecture for vision transformers with group-level parallelism, 2025. URL <https://arxiv.org/abs/2511.15397>.
- [31] Shiwei Liu, Chen Mu, Hao Jiang, Yunzhengmao Wang, Jinshan Zhang, Feng Lin, Keji Zhou, Qi Liu, and Chixiao Chen. HARDSEA: Hybrid analog-ReRAM clustering and digital-SRAM in-memory computing accelerator for dynamic sparse self-attention in transformer. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 32(2):269–282, 2024. doi: 10.1109/TVLSI.2023.3337777. URL <https://doi.org/10.1109/TVLSI.2023.3337777>.
- [32] W. Wan, R. Kubendran, B. Gao, S. Joshi, P. Raina, H. Wu, G. Cauwenberghs, and H.-S. P. Wong. A voltage-mode sensing scheme with differential-row weight mapping for energy-efficient RRAM-based in-memory computing. In *2020 IEEE Symposium on VLSI Technology*, pages 1–2, 2020. doi: 10.1109/VLSITechnology18217.2020.9265066. URL <https://doi.org/10.1109/VLSITechnology18217.2020.9265066>.
- [33] Xianggao Wang, Na Wei, Shifan Gao, Wenhao Wu, and Yi Zhao. Accuracy improvement with weight mapping strategy and output transformation for STT-MRAM-based computing-in-memory. *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, 10:75–81, 2024. doi: 10.1109/JXCDC.2024.3478360. URL <https://doi.org/10.1109/JXCDC.2024.3478360>.
- [34] Tomas Vincze, Michal Hanic, Martin Berki, and Martin Weis. Simultaneous dual-plasticity organic synaptic transistor for neuromorphic computing. *Advanced Electronic Materials*, 12(1):e00515, 2026. doi: 10.1002/aelm.202500515. URL <https://doi.org/10.1002/aelm.202500515>. Published online 2025-12-03.

Data Availability

All datasets used in this study (CIFAR-10, CIFAR-100, and Flowers-102) are publicly available through standard PyTorch torchvision and HuggingFace dataset interfaces. The source data underlying the main-text and Supplementary figures and tables are packaged with the submission bundle together with the corresponding manifests, logs, and figure-generation inputs. A submission-matched reproducibility archive containing the code snapshot, source data, SHA-256 manifest, and CITATION.cff file accompanies the reviewer materials; the same archive is staged for Zenodo deposition, with the DOI field reserved for acceptance-stage registration.

Code Availability

The reviewer-accessible archive contains the simulation framework, training and evaluation scripts, device-profile utilities, plotting code, and submission-matched source data needed to reproduce the figures and

tables. The codebase is distributed under the MIT licence, and the curated public repository is staged at <https://github.com/Leslie360/HAT>; Zenodo registration is reserved for the acceptance-stage public release.

Competing Interests

The authors declare no competing interests.